Linear programming: exercises

A15-L01 Free variables and minimization

Convert `min 3u+v; u−v=2; u+v≤6; v≥0; u free` to our standard and slack forms. Give both solution mappings, find the original optimum, and explain why simply adding $u \geq 0$ is not a general conversion.

A15-L02 Complete pivots and a dual certificate

Solve `max 4x₁+3x₂; 2x₁+x₂≤10; x₁+2x₂≤8; x≥0`, entering x_1 first. Give both complete dictionaries, each row's ratio, and all nonbasic coefficients. Construct the dual and verify feasibility and equal objectives.

A15-L03 Zero ratios, ties, and symbolic perturbation

For `x₃=−x₁; x₄=2−x₁−x₂; max x₁+x₂`, enter x_1, give the legal leaving row and zero-step dictionary, then finish solving. Separately, for `x₃=2−x₁; x₄=2−x₁−x₂`, give the lexicographic leaving row when $\text{RHS}=(2+\varepsilon, 2+\varepsilon^2)$ and x_1 enters.

A15-L04 From Phase I through the original optimum

Use shared artificial $a \geq 0$ for `max x; x≤4; x≥2; x≥0`. Give the infeasible initial slack point, the special-pivot feasible auxiliary dictionary, the auxiliary optimum, artificial removal, original-objective restoration, and Phase II. Change $x \leq 4$ to $x \leq 1$, find auxiliary optimum, and give an independent infeasibility certificate.

A15-L05 Improving rays and unbounded regions

Solve `max 2x₁+x₂; x₁−x₂≤1; x≥0`, entering x_1 first. Give the pivoted dictionary, a feasible x^o, and nonnegative direction r with $Ar \leq 0$, $c^T r > 0$. Prove every $x^o + \theta r, \theta \geq 0$, is feasible. Change objective to $-x_1 - x_2$ and explain the finite optimum on the same unbounded region.

A15-L06 Preconditions of the potential model

Graph edges are $s \to a: -2$, $a \to t: 3$, $s \to t: 4$, and a disconnected pair $u \to v: -1$, $v \to u: -1$. Write the s-to-t potential LP. Why is retaining all edges infeasible without making the actual s-to-t shortest distance nonfinite? Repair the model, solve it, and assess whether adding $d \geq 0$ is equivalent.

A15-L07 Shared capacities and integrality

Two commodities each demand 1 from s to t. Each can use two branches, each branch consisting of two capacity 1 arcs. Write the commodity LP and give fractional and integral feasible solutions. Then give a unique shared capacity 1 bottleneck with demands 3/4 each:prove individual feasibility but joint infeasibility. Distinguish fractional feasibility, existence of an integer solution, and requiring one path per commodity.

A15-L08 Complementary slackness and theorem boundaries

For $\max\ 2x_1+x_2;\ x_1+x_2\le 4;\ x_1\le 2;\ x\ge 0$, check x=(2,2), dual y=(1,1), and every gap product, proving optimality. For x=(1,3), identify the nonzero complementary product producing objective gap 1. Why does $\max\ x;\ 0\le x<1$ fall outside the LP trichotomy theorem?

A15-L09 Artificial variable and original objective

Use two phases to solve $\min\ 2x_1+x_2, x_1+x_2-x_3=5, x\ge 0$. Show the pivot taking the artificial variable from 5 to 0, remove its column, restore the original objective, and optimize.

A15-L10 Weak duality and complementary slackness

For $\max\ 3x_1+2x_2, x_1+2x_2\le 8,\ 3x_1+x_2\le 9,\ x\ge 0$, write the dual, prove weak duality and the gap decomposition, and certify $(2,3)$ with a dual candidate.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd edition, Chapter 29. MIT 6.046J Fall 2004, Lecture 15; Fall 2005, Lecture 18.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study